

Exercise 1. We say that the topological space (X, τ) is *metrizable* if there exists a metric d on X which induces the same topology as τ .

a) Show that if (X, τ) is metrizable then for each $x \in X$ there is a countable family $(V_n)_{n \in \mathbb{N}}$ of neighborhoods of x such that each open set U containing x , must contain a set V_n for some $n \in \mathbb{N}$.

b) Show that for any uncountable family $(X_i)_{i \in I}$ of non-void topological spaces, the space $\prod_{i \in I} X_i$ with the usual product topology is not metrizable.

Exercise 2. Show that for any countable family $(X_n)_{n \in \mathbb{N}}$ of metric spaces the space $\prod_{n \in \mathbb{N}} X_n$ with the usual product topology is metrizable.

Exercise 3. Let $(X_n)_{n \in \mathbb{N}}$ be a countable family of compact metric spaces. Prove without using Tychonov's theorem that $\prod_{n \in \mathbb{N}} X_n$ with the usual product topology is compact.

Exercise 4. Given two sets X, Y , let $\mathcal{F}(X, Y)$ denote the set of all functions $f : X \rightarrow Y$. Prove that a sequence (f_n) , $f_n : X \rightarrow \mathbb{R}$ converges in the product topology (in $\prod_{x \in X} \mathbb{R}$) if and only if it converges pointwise on X .

Exercise 5. Solve Exercise 3 on page 7 of the topology compendium.

Exercise 6. Prove Theorem 1.3 on page 3 of the topology compendium.